

Adriano Jucan

linkedin.com/in/adrianojucan | github.com/AdrianoJuc13

EDUCATION

University of Essex <i>Bachelor of Engineering, Robotic Engineering</i>	Colchester, UK Oct. 2019 – July 2024
National College "Petru Rares" <i>Mathematics and Informatics</i>	Suceava, RO Aug. 2015 – May 2019

PROJECTS

Developed a Data Glove System <i>ESP32, Unity, C#, Blender, IMU, Flex Sensors</i>	Oct 2023 – June 2024
- Engineered a data glove equipped with 5 flex sensors and 9-axis IMU sensors to accurately capture and augment user hand movements in real-time. - Integrated with Unity to animate a rigged virtual hand model, enhancing VR/AR interaction capabilities. - Achieved seamless real-time motion tracking, contributing to more immersive virtual environments.	
RC Flying Wing Plane <i>C++, nRF24, Arduino Nano</i>	Feb 2021 – June 2021
- Engineered a remote-controlled foam plane using nRF24 antennas, servos, Arduino Nano, and brushless motors, designed for long-range and stable flight performance. - Programmed flight control algorithms in C++ to optimize responsiveness and maneuverability, achieving reliable communication and control over extended distances. - Integrated locally sourced materials to create a lightweight yet durable structure, enhancing cost-efficiency and sustainability.	
Animated LED Panel for a Local Pub <i>Arduino Uno, C++, WS2811 LED Strip</i>	Dec 2021
- Designed and implemented a 6x5 meter animated LED ceiling panel using 300 meters of WS2811 LED strips and controlled by an Arduino Uno, creating a dynamic and visually engaging atmosphere. - Developed custom animations in C++, to enhance the pub's ambiance and customer experience. - Optimized power distribution and control circuits to ensure consistent performance and reliability, managing high power loads across multiple LED strips.	
Robotic Arm <i>Arduino Uno, Servo Motors, C++</i>	2017 – 2018
Constructed a functional robotic arm prototype using 5 servo motors, strings, and cardboard, demonstrating basic robotic movement and control.	

EXPERIENCE

Programming Assistant — Intern <i>Primagra</i>	May 2022 – Oct 2022 Suceava, RO
Developed Python software for data management, built a SWAT Risk Management App with React.js for improved risk analysis, and created catalogues and presentations.	
First Tech Challenge — Designer, Robot Driver <i>National College "Petru Rares"</i>	2018 – 2019 Suceava, RO
3D Designer of the robot using Solidworks - Robot team driver - Team collaboration on project planning, and tasks that required creativity.	

SKILLS

Programming Languages: Python, C/C++, Java, C#, SQL, JavaScript, Dart
Embedded Systems & Robotics: Arduino, ESP32, ROS2, OpenCV, NumPy, Matplotlib, nRF24, I2C, SPI, UART
Software & Tools: VS Code, PyCharm, IntelliJ, Git, GitHub, Firebase, Jupyter
3D Modeling & Simulation: SolidWorks, Blender, Unity, 3D Printing
Languages: English (Advanced), Romanian (Native), German (Beginner), Japanese (Beginner)